

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(GP_FFCutsFits);
nf = 11

```

```

nf =
11

```

```

%for i = 11:numel(fn)
    %selectCuts = fn{nf};
    selectCuts = 'Ei400_off200GPmaxFF'

```

```

selectCuts =
'Ei400_off200GPmaxFF'

```

```

if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei400_off200GPmaxFF

```

```

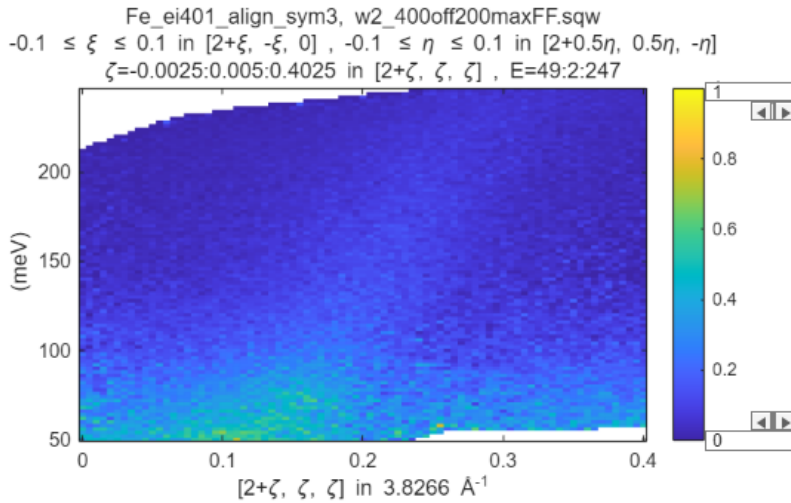
%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = GP_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample = IX_sample(true, [1, 0, 0], [0, 1, 0], 'cuboid', [0.04, 0.03, 0.02]);
for j = 1:numel(cuts_list)

```

```

cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
end

```



```

dir_name = "GP";

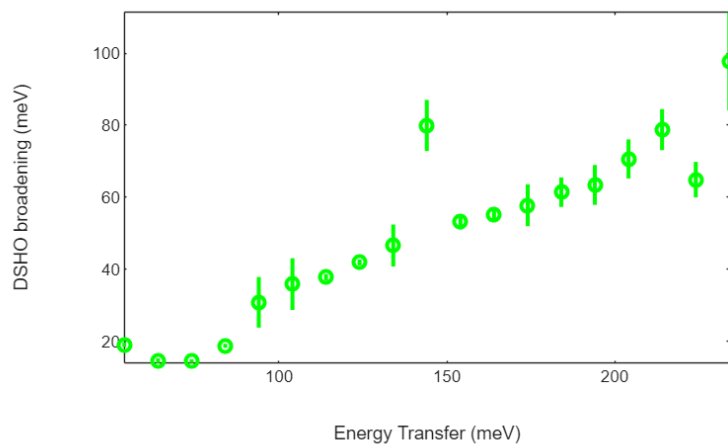
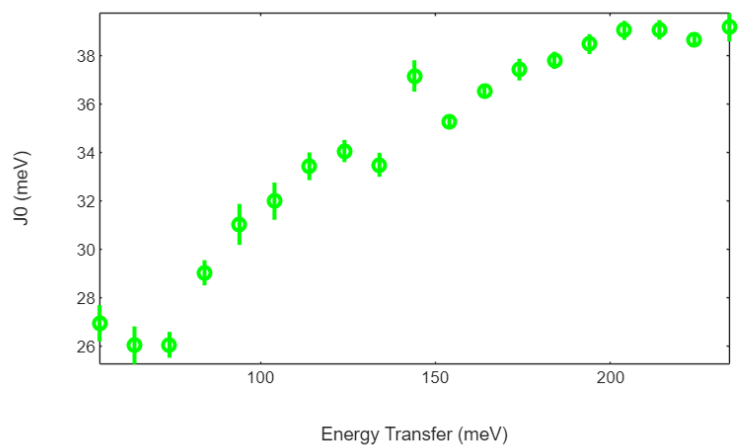
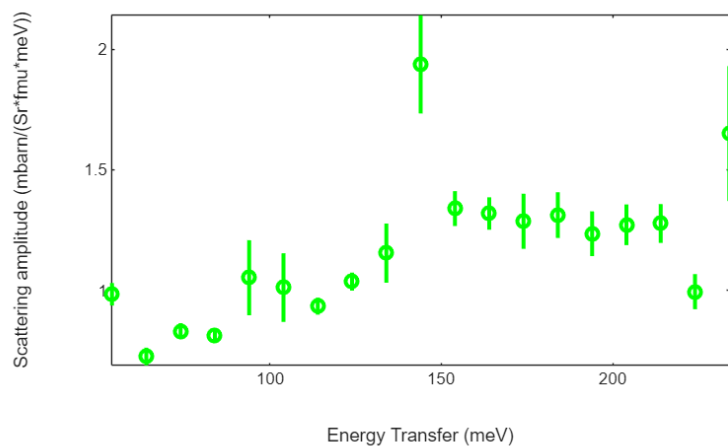
dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

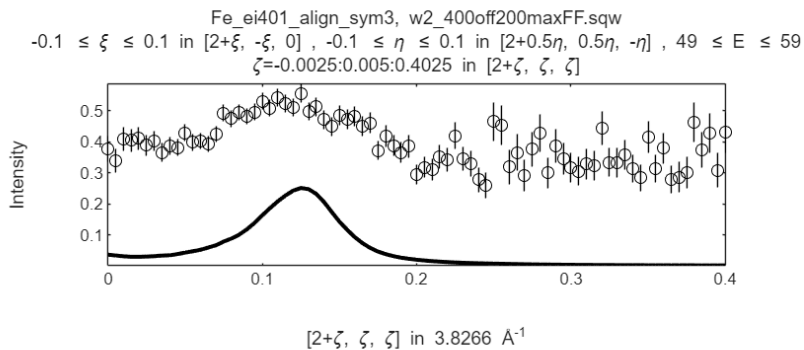
cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);

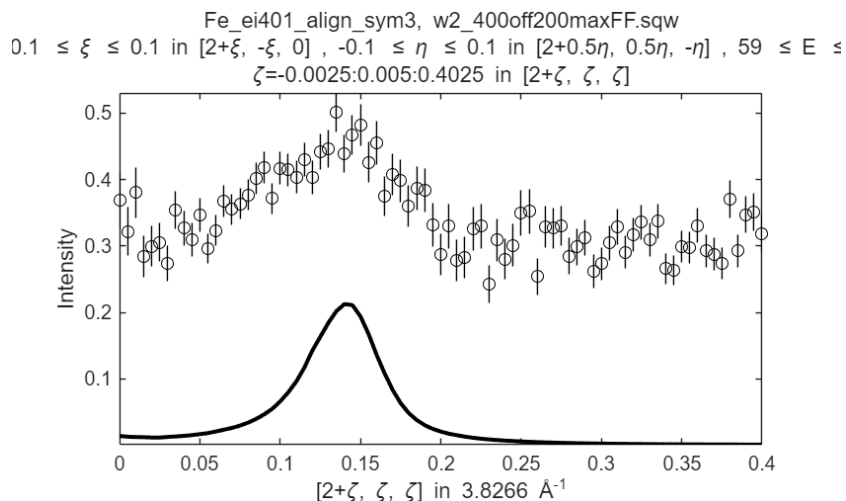
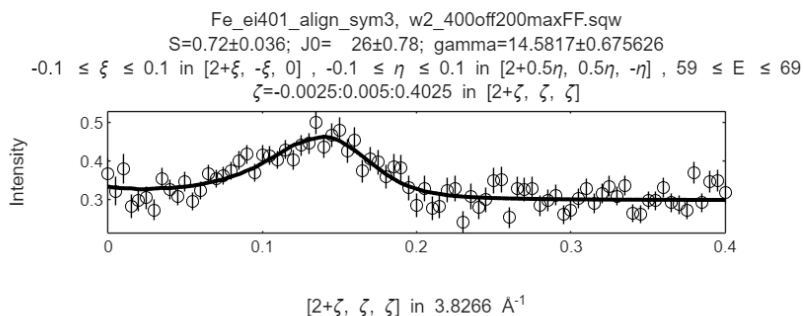
```



 **** En = 54±5 going to 234

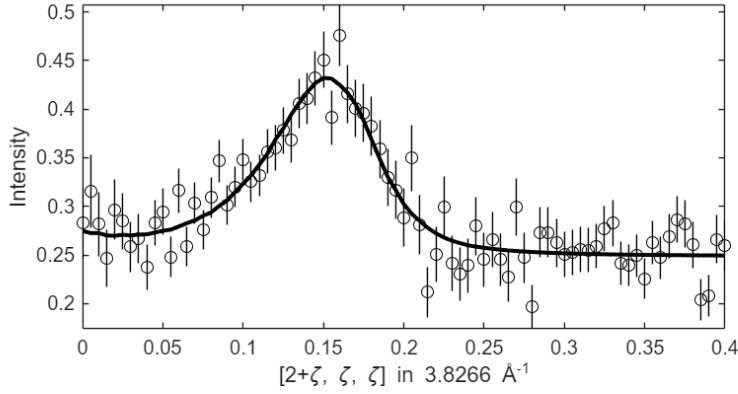


 **** En = 64 ± 5 going to 234

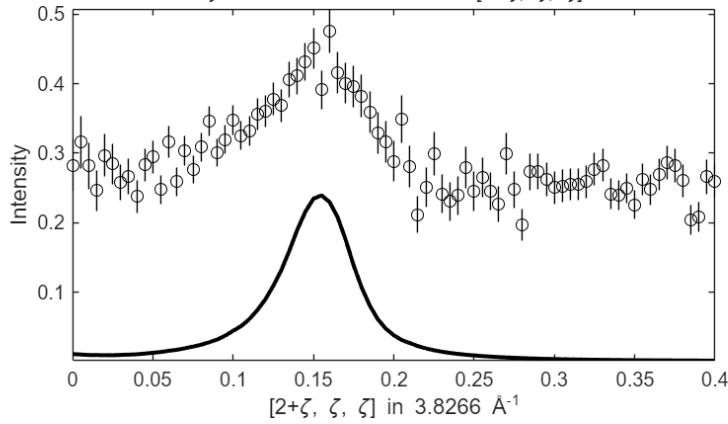


 **** En = 74 ± 5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.83\pm0.088$; $J0= 26\pm0.55$; $\gamma=14.5136\pm3.368$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $69 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

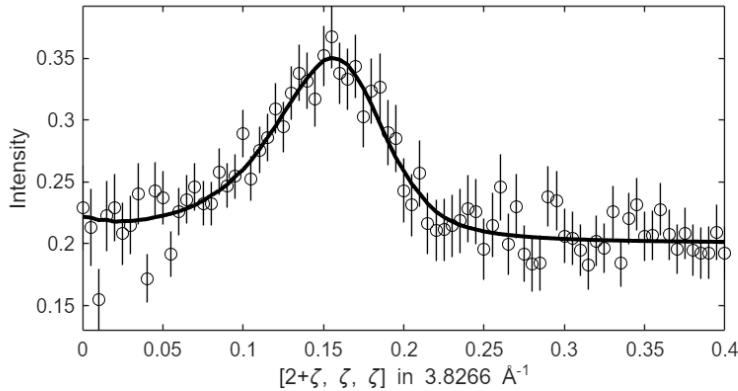


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.83\pm0.088$; $J0= 26\pm0.55$; $\gamma=14.5136\pm3.368$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $69 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

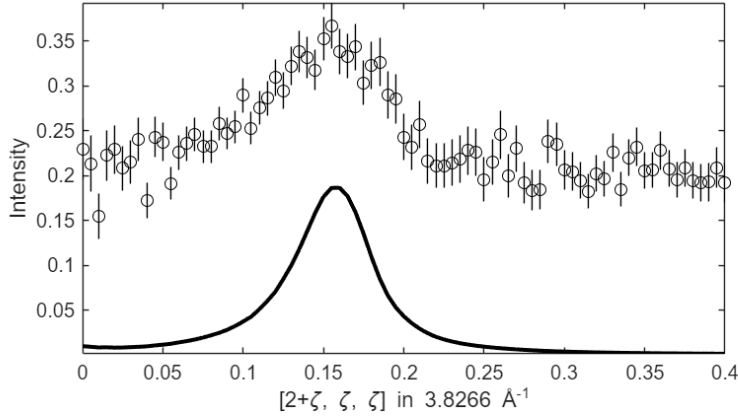


**** En = 84 ± 5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.82\pm0.029$; $J0= 29\pm0.56$; $\gamma=18.9215\pm0.543646$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $79 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

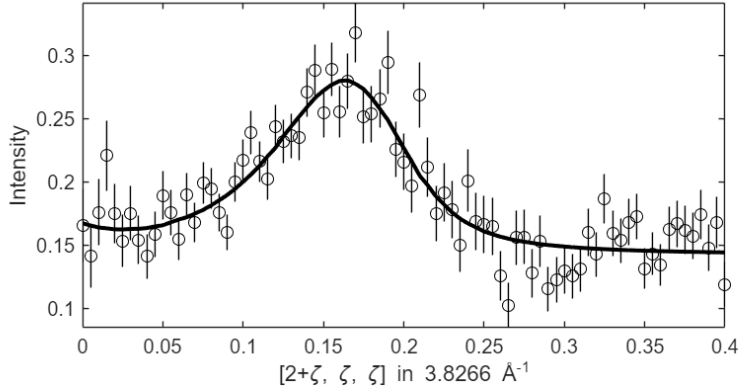


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $79 \leq E \leq 89$ eV
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

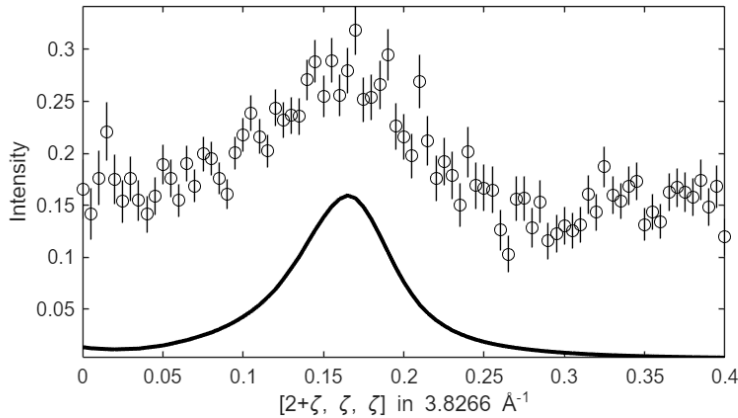


**** En = 94 ± 5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1 \pm 0.046$; $J_0 = 31 \pm 0.72$; $\gamma = 30.8594 \pm 0.953694$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $89 \leq E \leq 89$ eV
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

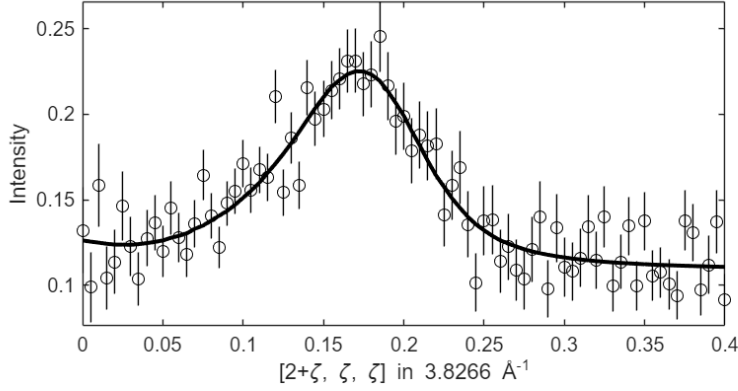


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $89 \leq E \leq 89$ eV
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

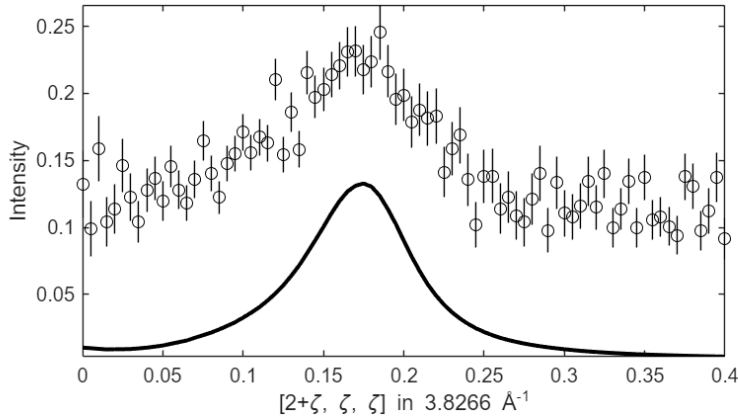


**** En = 104 ± 5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=1\pm0.14$; $J0=32\pm0.76$; $\gamma=35.2021\pm7.03029$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $99 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

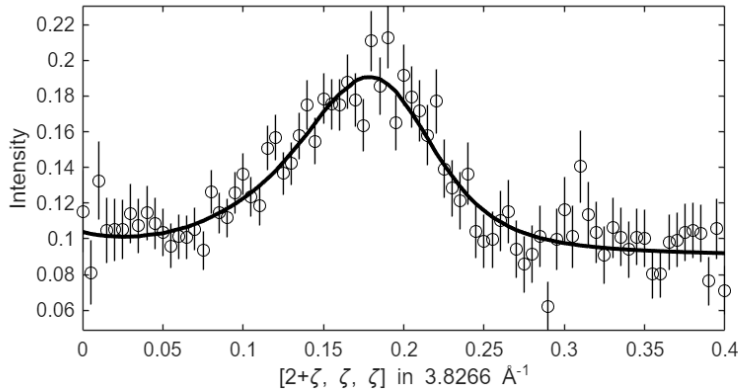


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $99 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

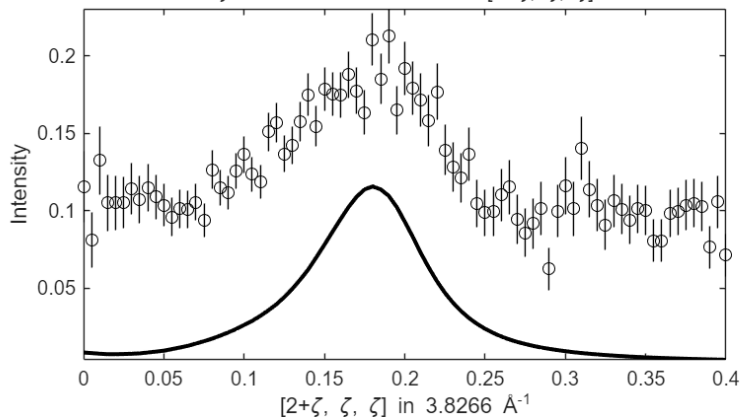


**** En = 114±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.93\pm0.036$; $J0=33\pm0.58$; $\gamma=37.7778\pm0.88747$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $109 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

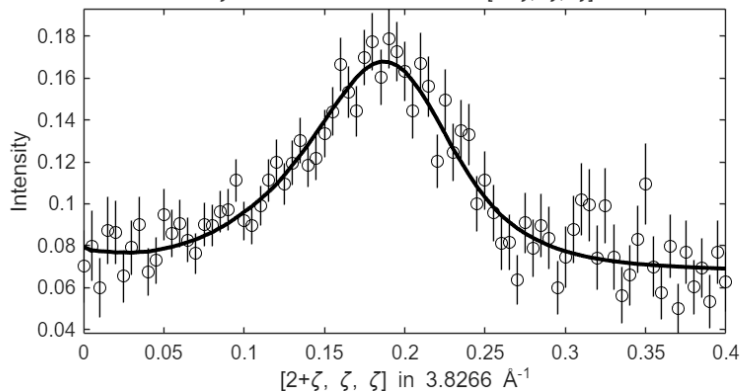


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $109 \leq E \leq 119$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

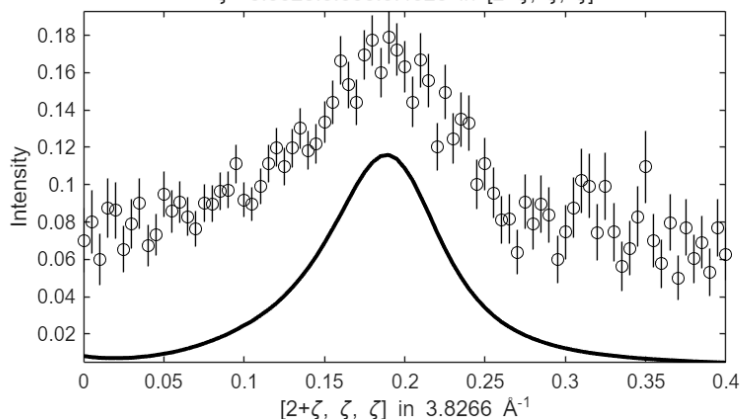


**** En = 124±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1 \pm 0.037$; $J_0 = 34 \pm 0.46$; $\gamma = 41.9022 \pm 0.814696$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $119 \leq E \leq 129$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

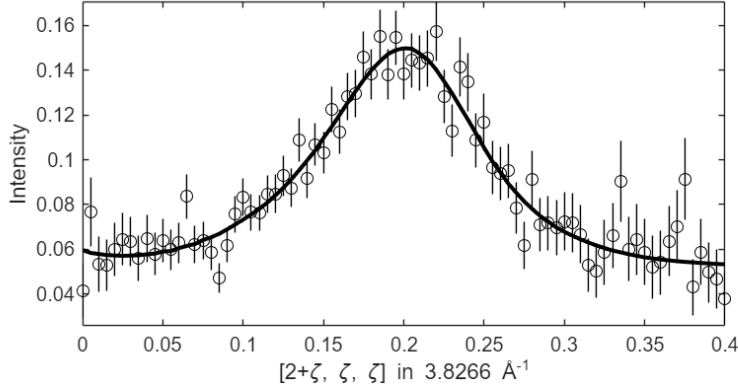


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $119 \leq E \leq 129$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

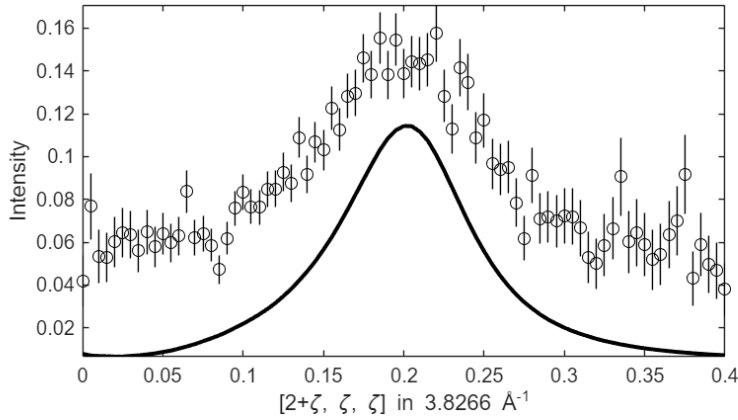


**** En = 134±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.2 \pm 0.12$; $J_0 = 34 \pm 0.5$; $\gamma = 46.4798 \pm 5.82561$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $129 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

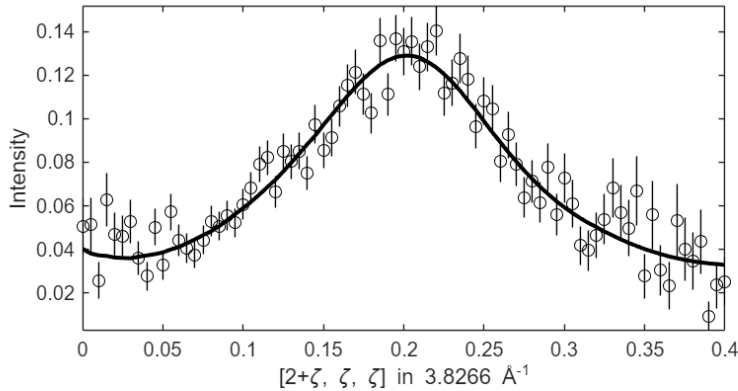


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.2 \pm 0.12$; $J_0 = 34 \pm 0.5$; $\gamma = 46.4798 \pm 5.82561$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $129 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

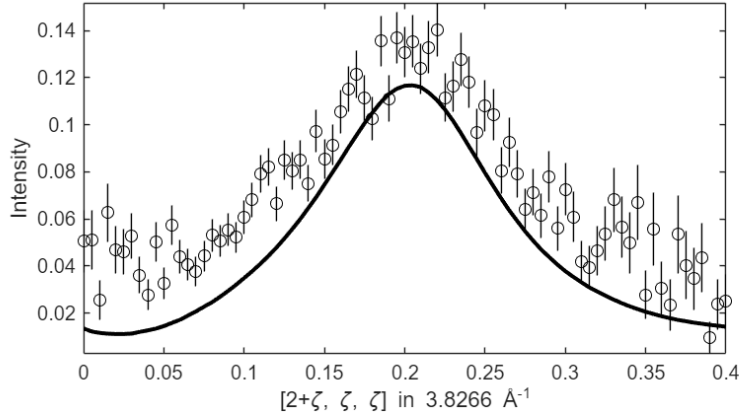


 **** En = 144±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.9 \pm 0.28$; $J_0 = 37 \pm 1.1$; $\gamma = 79.6522 \pm 12.4546$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $139 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

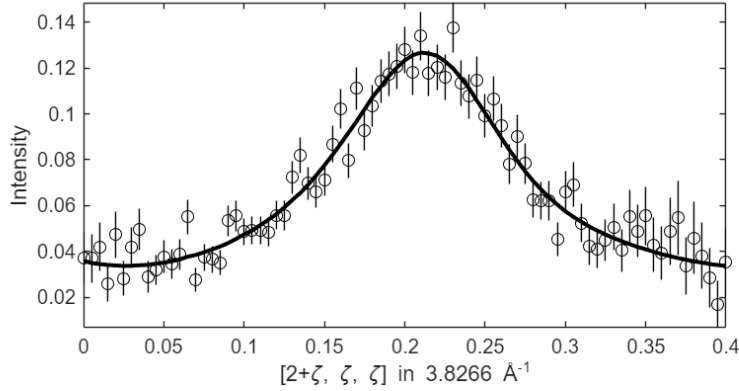


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $139 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

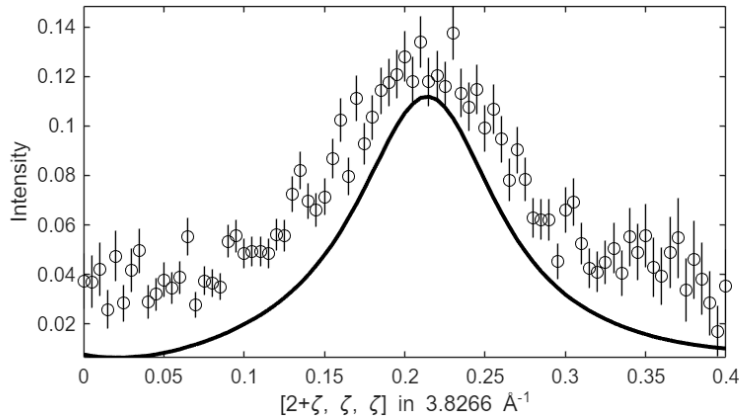


 **** En = 154±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.4 \pm 0.12$; $J_0 = 35 \pm 0.45$; $\gamma = 54.135 \pm 5.56541$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $149 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

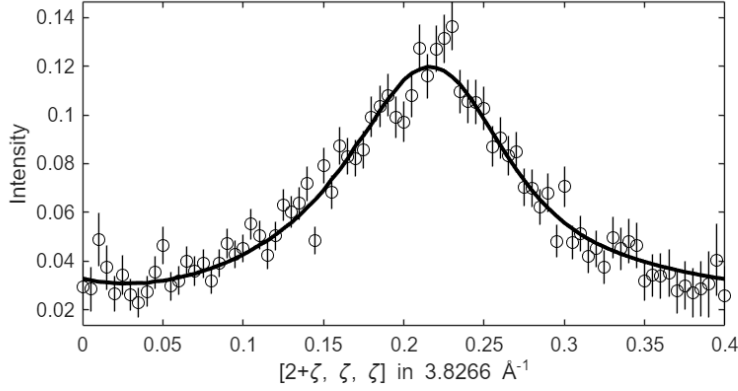


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $149 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

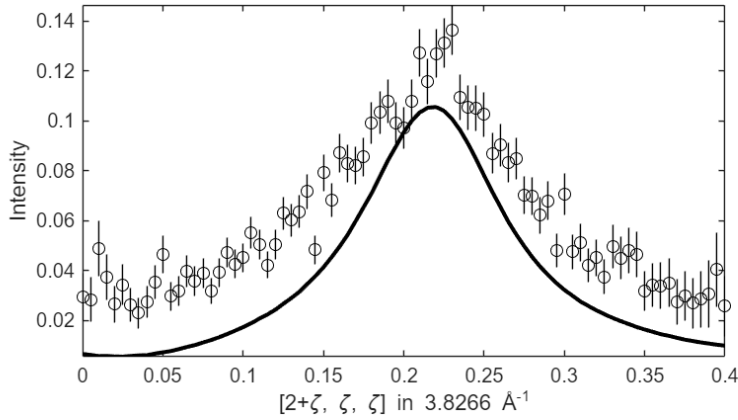


 **** En = 164±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.11$; $J_0 = 37 \pm 0.41$; $\gamma = 55.3758 \pm 5.19009$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $159 \leq E \leq 169$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

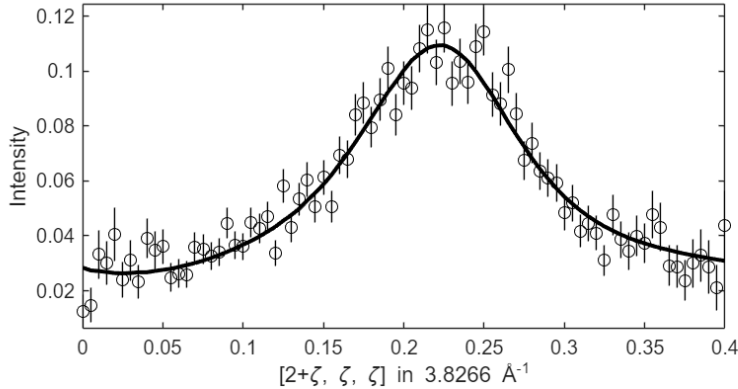


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.11$; $J_0 = 37 \pm 0.41$; $\gamma = 55.3758 \pm 5.19009$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $159 \leq E \leq 169$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

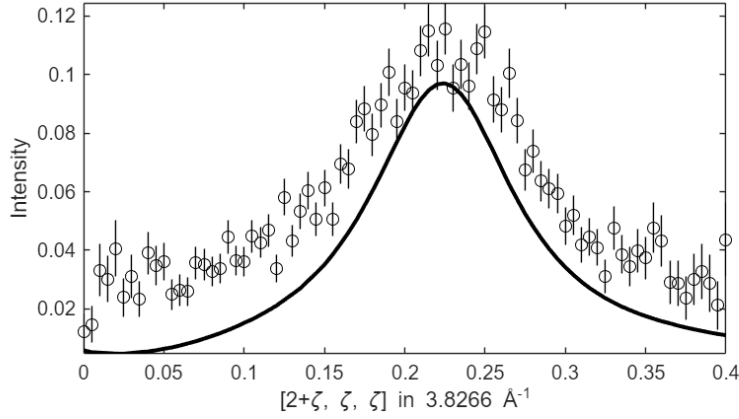


 **** En = 174±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.078$; $J_0 = 37 \pm 0.32$; $\gamma = 57.8014 \pm 2.81342$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $169 \leq E \leq 179$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

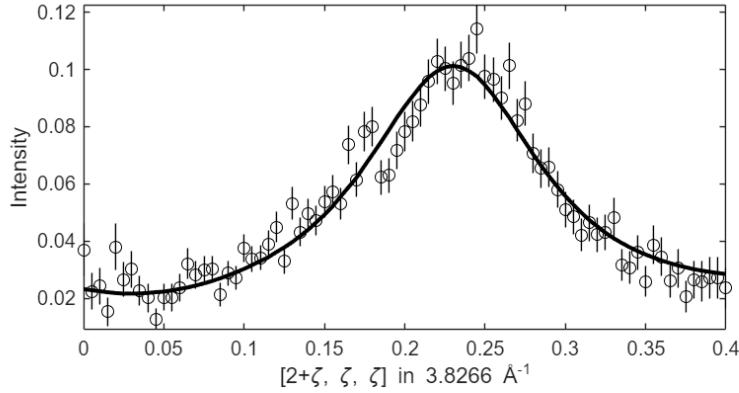


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $169 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

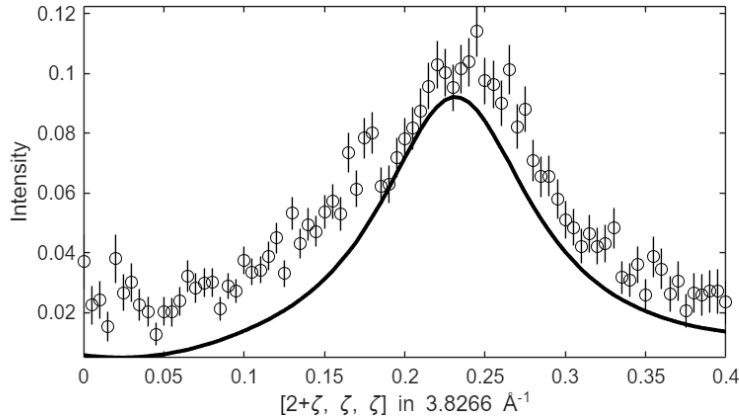


 **** En = 184±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 S= 1.3±0.11; J0= 38±0.44; gamma=61.4482±5.88261
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $179 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

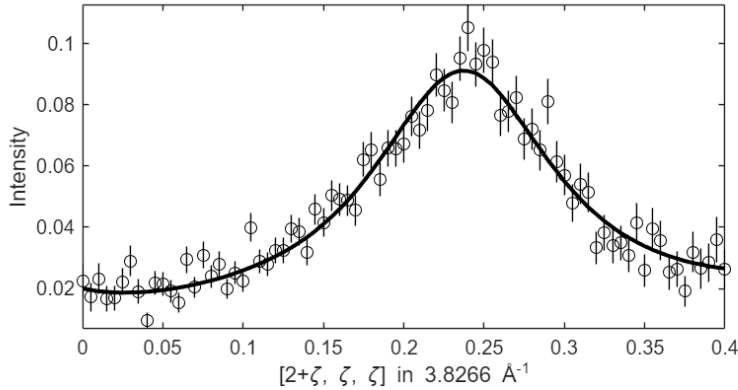


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $179 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$



 **** En = 194±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.2 \pm 0.093$; $J_0 = 38 \pm 0.41$; $\gamma = 63.1945 \pm 5.50851$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $189 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$



**** En = 204±5 going to 234

**** En = 214±5 going to 234

**** En = 224±5 going to 234

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 4.9383

**** En = 234±5 going to 234

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.4691

```
c1 = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [c1{:}];
```

```
beep
beep
beep
```

```
all_fit_par = GP_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```

```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
```

```
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
```

```
export(thisFile,targ_file);
```

```
Error using export>checkWritePermission (line 251)
Unable to overwrite existing file. Check write permission or whether the file is open in another application.

Error in export (line 104)
    checkWritePermission(results.Destination);
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
close all;  
%end
```

```
fit_src_struct = GP_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```